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INLET GUIDE VANE AND ROTARY MACHINE

Abstract

An inlet guide vane according to the disclosure includes a plurality of movable blades disposed at intervals in a circumferential direction of a rotor shaft and rotatable about a center axis extending in a radial direction of the rotor shaft. Each of the movable blades includes a first blade surface and a second blade surface extending in a direction connecting a leading edge and a trailing edge, and an inclined surface formed at a trailing end including the trailing edge and extending obliquely from the first blade surface to the second blade surface when viewed from the radial direction. The inclined surface is a flat surface whose cross-sectional shape viewed from the radial direction is inclined so as to be connected, at an acute angle, to the second blade surface at the trailing edge.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to Japanese Patent Application Number 2024-025098 filed on Feb. 22, 2024. The entire contents of the above-identified application are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The disclosure relates to an inlet guide vane and a rotary machine.

RELATED ART

[0003] A centrifugal compressor causes a working fluid to flow inside a rotating impeller, and compresses the working fluid in a gas state using a centrifugal force generated when the impeller rotates. As disclosed in JP 2016-200119 A, some of such centrifugal compressors have a structure in which an inlet guide vane is provided so as to adjust the flow rate of a working fluid introduced from the outside.

[0004] JP 2016-200119 A describes an inlet guide vane including a plate portion having a plate-like shape in which movable blades are disposed in a flow path. In a cross section orthogonal to a rotor shaft, the plate portion includes a bent portion inclined with respect to other portions at a leading end of the flow path on its upstream side. This enables air to flow into the flow path efficiently with a simple shape.

SUMMARY

[0005] In the inlet guide vane described above, a Karman vortex may be generated on the downstream side relative to the movable blades. When a Karman vortex is generated, the flow of the fluid having passed through the inlet guide vane is disturbed, which may not only cause a pressure loss, but also exert adverse effects such as noises and vibrations. Therefore, it is desired to suppress the generation of a Karman vortex due to the fluid having passed through the inlet guide vane.

[0006] The disclosure has been made to solve the above problems, and an object of the disclosure is to provide an inlet guide vane and a rotary machine that can suppress generation of a Karman vortex due to a fluid having passed through the inlet guide vane.

[0007] In order to solve the above problems, an inlet guide vane according to the disclosure includes a plurality of movable blades disposed at intervals in a circumferential direction of a rotor shaft and rotatable about a center axis extending in a radial direction of the rotor shaft. Each of the movable blades includes a first blade surface and a second blade surface extending in a direction connecting a leading edge and a trailing edge, and an inclined surface formed at a trailing end including the trailing edge and extending obliquely from the first blade surface to the second blade surface when viewed from the radial direction. The inclined surface is a flat surface whose cross-sectional shape viewed from the radial direction is inclined so as to be connected, at an acute angle, to the second blade surface at the trailing edge.

[0008] A rotary machine according to the disclosure includes: a rotor including a rotor shaft extending in an axial direction in which an axis extends with the axis as a center, and an impeller fixed to the rotor shaft; a casing covering the rotor and including a suction port through which a working fluid flows into the casing; and the inlet guide vane disposed in the casing on a first side in the axial direction with respect to the impeller.

[0009] In the inlet guide vane and the rotary machine of the disclosure, it is possible to suppress the generation of a Karman vortex due to a fluid having passed through the inlet guide vane.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] The disclosure will be described with reference to the accompanying drawings, wherein like numbers reference like elements.

[0011] FIG. 1 is a diagram illustrating a schematic configuration of a rotary machine according to an embodiment of the disclosure.

[0012] FIG. 2 is a cross-sectional view illustrating a configuration of the rotary machine in which movable blades of an inlet guide vane are in a fully-opened state.

[0013] FIG. 3 is a schematic diagram illustrating the inlet guide vane as viewed from a radial direction.

[0014] FIG. 4 is a schematic diagram illustrating an inlet guide vane of a modification example as viewed from the radial direction.

DESCRIPTION OF EMBODIMENTS

[0015] Hereinafter, embodiments for implementing an inlet guide vane and a rotary machine according to the disclosure will be described with reference to the accompanying drawings.

However, the disclosure is not limited only to these embodiments.

Embodiments

Configuration of Geared Compressor

[0016] As illustrated in FIGS. 1 and 2, a geared compressor (centrifugal compressor) 1 serving as a rotary machine according to the present embodiment mainly includes a rotor 3, a casing 2 (see FIG. 2), an inlet guide vane 6 (see FIG. 2), a radial bearing 12, and a thrust bearing 17.

Configuration of Rotor

[0017] The rotor 3 is rotatable about an axis O relative to the casing 2. The rotor 3 includes a rotor shaft 30, an impeller 40, and an impeller cap 38.

[0018] The rotor shaft 30 extends in an axial direction Da in which an axis O extends with the axis O as a center. As illustrated in FIG. 1, the rotor shaft 30 is supported by a pair of radial bearings 12 so as to be rotatable about the axis O. The pair of radial bearings 12 is disposed at an interval in the axial direction Da. The rotor shaft 30 is constrained from moving in the axial direction Da by a pair of thrust bearings 17. The pair of thrust bearings 17 is disposed between the pair of radial bearings 12 at positions spaced apart from each other on both sides of a pinion gear 15 described later in the axial direction Da.

[0019] The rotor shaft 30 is connected to an external drive source (not illustrated) such as a motor via a speed increase transmitter 11. The speed increase transmitter 11 includes the pinion gear 15 and a large-diameter gear 16. The pinion gear 15 is fixed to the rotor shaft 30 between the pair of radial bearings 12. The large-diameter gear 16 engages with the pinion gear 15. The large-diameter gear 16 is rotationally driven by the drive source. The large-diameter gear 16 is set to have a larger outside diameter dimension than that of the pinion gear 15. Thus, the rotational speed of the rotor shaft 30 having the pinion gear 15 fixed thereto is higher than the rotational speed of the large-diameter gear 16. That is, the speed increase transmitter 11 increases, via the pinion gear 15, the rotational speed of the large-diameter gear 16 rotated by the external drive source, and transmits the increased rotational speed to the rotor shaft 30.

[0020] Impellers 40 are disposed at opposite end portions of the rotor shaft 30 in the axial direction Da. In the present embodiment, each of the impellers 40 is a so-called closed impeller including a disc 41, a blade 42, and a cover 43 as illustrated in FIG. 2. Note that the impeller 40 may be an open impeller without the cover 43.

[0021] The disc 41 has a disc shape and is fixed to the rotor shaft 30. The disc 41 includes a first surface 41a facing the cover 43 in the axial direction Da and a second surface 41b facing the opposite side to the first surface 41a in the axial direction Da. The second surface 41b is a back

surface of the impeller **40**. Here, in the present embodiment, the geared compressor **1** includes one impeller **40** at each of both end portions of the rotor shaft **30** in the axial direction Da as illustrated in FIG. **1**. Each of the impellers **40** is disposed in such a manner that, in the axial direction Da , the second surface **41b** of the disc **41**, which is a back surface, faces the pinion gear **15** and the first surface **41a** faces the end portion of the rotor shaft **30** on the side opposite to the pinion gear **15**. That is, a first stage impeller **40A** provided at a first end of the rotor shaft **30** and a second stage impeller **40B** provided at a second end of the rotor shaft **30** are disposed with the discs **41** facing opposite to each other in the axial direction Da such that their back surfaces face each other.

[0022] In the following description, in each of the impellers **40**, a side of the disc **41** near the first surface **41a** is referred to as a first side $Da1$ in the axial direction Da , and another side of the disc **41** near the second surface **41b** is referred to as a second side $Da2$ in the axial direction Da . That is, the first side $Da1$ in the axial direction Da and the second side $Da2$ in the axial direction Da are inverse between the first stage impeller **40A** and the second stage impeller **40B**.

[0023] As illustrated in FIG. **2**, the blade **42** extends from the first surface **41a** of the disc **41** to the cover **43**. A plurality of blades **42** are disposed at intervals in a circumferential direction Dc of the rotor shaft **30** around the axis O .

[0024] The cover **43** is disposed on the first side $Da1$ in the axial direction Da with respect to the disc **41** and the plurality of blades **42**. The cover **43** has a disc shape and is formed to cover the plurality of blades **42**.

[0025] A working fluid (e.g., air) flows from the first side $Da1$ in the axial direction Da toward the second side $Da2$ in the axial direction Da with respect to the impeller **40**. In each of the impellers **40**, an impeller flow path **45** is formed between the disc **41** and the cover **43**. The impeller flow path **45** includes an inflow port **45i** and an outflow port **45o**. The inflow port **45i** is open to face the first side $Da1$ in the axial direction Da on an inner side Dri in a radial direction Dr in the impeller **40**. Here, the radial direction Dr is a radial direction Dr of the rotor shaft **30** centered on the axis O . The outflow port **45o** is open toward an outer side Dro in the radial direction Dr in the impeller **40**.

[0026] A shaft end **30s** which is an end portion of the rotor shaft **30** in the axial direction Da protrudes toward the first side $Da1$ in the axial direction Da with respect to the impeller **40**. An impeller cap **38** is fixed to the shaft end **30s**. The impeller cap **38** rotates together with the rotor shaft **30**. The impeller cap **38** forms a rotor end portion **3e** which is an end portion of the rotor **3** in the axial direction Da . The impeller cap **38** restricts the movement of the impeller **40** in the axial direction Da . That is, the impeller cap **38** constrains the position of the impeller **40** in the axial direction Da such that the impeller **40** does not fall off from the rotor shaft **30**.

Configuration of Casing

[0027] As illustrated in FIG. **2**, the casing **2** is formed to cover the rotor **3**. The casing **2** is made of metal and forms an outer shell of the geared compressor **1**. The casing **2** includes a shaft insertion hole **21**, into which the rotor shaft **30** is inserted, on the second side $Da2$ in the axial direction Da with respect to the position at which the impeller **40** is disposed. The casing **2** includes an intake nozzle **22** and an exhaust flow path **23** around each of the impellers **40**.

[0028] The intake nozzle **22** allows the working fluid to flow into the interior of the casing **2**. The intake nozzle **22** is formed in a tubular shape extending in the axial direction Da . A suction port **22a** centered on the axis O is formed inside the intake nozzle **22**. The intake nozzle **22** communicates, through the suction port **22a**, with the outside of the casing **2** and with the inflow port **45i** of the impeller flow path **45** opened to the inner side Dri in the radial direction Dr in the impeller **40**. When the impeller **40** rotates in the circumferential direction Dc around the axis O , the working fluid is sucked from the outside into the interior of the casing **2** through the suction port **22a**.

[0029] The exhaust flow path **23** allows the working fluid inside the casing **2** to flow out of the casing **2**. The exhaust flow path **23** is formed on the outer side Dro in the radial direction Dr relative to the outflow port **45o** of the impeller flow path **45**. The exhaust flow path **23** has a spiral shape continuous in the circumferential direction Dc .

Configuration of Inlet Guide Vane

[0030] The inlet guide vane **6** controls the flow rate of the working fluid passing through the suction port **22a**. The inlet guide vane **6** is disposed on the inner side Dri of the intake nozzle **22** of the casing **2**. That is, the inlet guide vane **6** is disposed on the first side Da1 in the axial direction Da with respect to the impeller **40** in the casing **2**. The inlet guide vane **6** includes a plurality of movable blades **60**. The plurality of movable blades **60** are disposed so as to protrude into the suction port **22a** having a circular cross section when viewed from the axial direction Da. The plurality of movable blades **60** are disposed at equal intervals in the circumferential direction Dc around the axis O along the inside surface of the intake nozzle **22**.

[0031] Each of the movable blades **60** is rotatable about a center axis Ar extending in the radial direction Dr. As illustrated in FIG. 3, the plurality of movable blades **60** increase the opening degree by rotating toward the reverse side to a rotational direction R of the rotor shaft **30**. Here, the rotational direction R is a direction in which the rotor shaft **30** is directed from the rear side toward the front side in the circumferential direction Dc.

[0032] Here, a state in which a blade chord direction D2 of the movable blades **60** is parallel to a flow direction F (the axial direction Da) of the working fluid is referred to as a fully-opened state of the movable blades **60**. That is, the fully-opened state is a rotated state in which each of the movable blades **60** (blade main bodies **61**) becomes thickest in a cross-sectional view parallel to the axis O. When the movable blades **60** are brought into the fully-opened state, the flow rate of the working fluid passing through the suction port **22a** becomes 100%. When the movable blades **60** further rotate in the same direction as the rotational direction R of the rotor shaft **30** from the fully-opened state as viewed from the radial direction Dr, the flow rate of the working fluid passing through the suction port **22a** becomes 120%. A state in which the movable blades **60** are arranged in this manner is referred to as an over-opened state of the movable blades **60**. On the other hand, when each of the movable blades **60** is rotated around the center axis Ar from the fully-opened state toward the reverse side to the rotational direction R of the rotor shaft **30** as viewed from the radial direction Dr such that the blade chord direction D2 intersects the flow direction F (the axial direction Da) of the working fluid, then, the suction port **22a** is gradually closed by the blade main bodies **61**. As a result, the flow rate of the working fluid flowing from the suction port **22a** into the impeller **40** through the inlet guide vane **6** is reduced. A state in which the blade chord direction D2 is orthogonal to the flow direction F (the axial direction Da) of the working fluid is referred to as a fully-closed state of the movable blades **60**. That is, the fully-closed state is a rotated state in which each of the movable blades **60** (the blade main bodies **61**) becomes thinnest in a cross-sectional view parallel to the axis O. As illustrated in FIGS. 2 and 3, in the present embodiment, each of the movable blades **60** includes the blade main body **61** and a shaft portion **62**.

[0033] Each blade main body **61** extends so as to protrude from the inside surface of the intake nozzle **22** in a blade height direction D1 which is a direction (the radial direction Dr) in which the center axis Ar extends. Here, a direction connecting a leading edge **61f** and a trailing edge **61r** of the blade main body **61** having a blade cross-sectional shape, and orthogonal to the blade height direction D1 (the radial direction Dr) is referred to as the blade chord direction D2. The blade main body **61** includes a first blade surface **611**, a second blade surface **612**, and an inclined surface **613**.

[0034] The first blade surface **611** and the second blade surface **612** extend in the blade chord direction D2. Each of the first blade surface **611** and the second blade surface **612** is formed of a flat surface whose cross-sectional shape viewed from the radial direction Dr forms a straight line along the blade chord direction D2. That is, the first blade surface **611** and the second blade surface **612** are not formed with irregularities or curved surfaces when viewed from the radial direction Dr. Thus, the blade main body **61** has a flat plate-like shape extending in the blade chord direction D2 in cross section viewed from the radial direction Dr. The first blade surface **611** and the second blade surface **612** are parallel to each other. In the blade chord direction D2, the second blade surface **612** is formed to be longer than the first blade surface **611**. The first blade surface **611** is

located in front of the second blade surface **612** in the rotational direction R of the rotor shaft **30**. [0035] The inclined surface **613** is formed at a trailing end including the trailing edge **61r**. The inclined surface **613** extends obliquely from the first blade surface **611** to the second blade surface **612**. The inclined surface **613** is a flat surface whose cross-sectional shape viewed from the radial direction Dr is inclined so as to be connected, at an acute angle, to the second blade surface **612** at the trailing edge **61r**. Thus, the inclined surface **613** is inclined to face the first side Da1 of the axial direction Da so as to be opposed to the suction port **22a** when the movable blade **60** is in the fully-closed state. The inclined surface **613** is formed as a flat surface forming a straight line without irregularities or curved surfaces when viewed from the radial direction Dr. The angle of the inclined surface **613** with respect to the first surface **41a** and the second surface **41b** is constant in the blade height direction D1. The inclined surface **613** is inclined at an angle of 15° or more and 45° or less with respect to the second blade surface **612** when viewed from the radial direction Dr. Preferably, the inclined surface **613** is inclined at an angle of 25° or more and 35° or less with respect to the second blade surface **612** when viewed from the radial direction Dr. The trailing end at which the inclined surface **613** is formed is a partial region including the trailing edge **61r** located away from the center axis Ar when the movable blade **60** is viewed from the radial direction Dr. The trailing end is a region of 1/3 or less, and in the present embodiment, a region of 1/4 or less, of the total length of the movable blade **60** in the blade chord direction D2.

[0036] As illustrated in FIG. 2, the shaft portion **62** is formed to protrude from the blade main body **61** toward the outer side Dro in the radial direction Dr. The shaft portion **62** is formed integrally with the blade main body **61**. The shaft portion **62** is inserted into a shaft support hole **22h** formed in the intake nozzle **22**. The shaft portion **62** is rotatable about the center axis Ar by a blade drive device (not illustrated) in a state of being inserted in the shaft support hole **22h**. Thus, the blade main body **61** is rotatable about the center axis Ar integrally with the shaft portion **62**. In each of the movable blades **60**, the angle of the blade main body **61** with respect to the flow direction F (the axial direction Da) of the working fluid flowing through the suction port **22a** is adjusted by rotating the blade main body **61** about the center axis Ar. The inlet guide vane **6** is opened and closed by rotating each of the plurality of movable blades **60** about the center axis Ar.

[0037] In the geared compressor **1** described above, when the impeller **40** rotates integrally with the rotor shaft **30**, the working fluid is sucked into the intake nozzle **22** of the casing **2** from the suction port **22a**. In the suction port **22a**, the flow rate of the working fluid is adjusted by the opening degree of the inlet guide vane **6** when the working fluid passes through the inlet guide vane **6**. The working fluid having passed through the inlet guide vane **6** is drawn into the impeller flow path **45** from the intake nozzle **22** through the inflow port **45i**. The working fluid flows from the inflow port **45i** toward the outflow port **45o** by a centrifugal force generated by the impeller **40** rotating integrally with the rotor shaft **30**. The working fluid is compressed while flowing from the inflow port **45i** toward the outflow port **45o**. The compressed working fluid flows out from the outflow port **45o** to the outer side Dro in the radial direction Dr to be fed to the exhaust flow path **23** on the outer side Dro in the radial direction Dr. The working fluid is further compressed while swirling around the axis O along the exhaust flow path **23**.

Operational Effects

[0038] In the inlet guide vane **6** and the geared compressor **1** described above, the inclined surface **613** is formed at the trailing end of the movable blade **60** so as to extend obliquely from the first blade surface **611** to the second blade surface **612** and be connected at an acute angle to the second blade surface **612** at the trailing edge **61r**. That is, the trailing end of the movable blade **60** has an asymmetrical shape due to the inclined surface **613** when viewed from the radial direction Dr. As a result, it is possible to suppress the generation of a Karman vortex due to a fluid having passed through the inlet guide vane **6**.

[0039] In addition, each of the first blade surface **611** and the second blade surface **612** is formed of a flat surface whose cross-sectional shape viewed from the radial direction Dr forms a straight line

along the blade chord direction D2. The first blade surface **611** and the second blade surface **612** are parallel to each other. That is, the movable blade **60** has a linear shape. Thus, when the inlet guide vane **6** is brought close to the fully-closed state, a gap between the movable blades **60** adjacent to each other can be reduced. Accordingly, the leakage amount of the working fluid flowing through the gap can be reduced. As a result, even when the opening degree of the inlet guide vane **6** is low, opening-degree characteristics can be improved.

[0040] In addition, since the inclined surface **613** is formed as a flat surface, the workability of the inclined surface **613** can be improved. As a result, it is possible to easily form such a shape as to suppress the generation of a Karman vortex at the trailing end of the movable blade **60**. Further, since not only the inclined surface **613**, but also the first blade surface **611** and the second blade surface **612** are formed as flat surfaces, the workability of the movable blade **60** in its entirety can be improved. As a result, not only the inclined surface **613**, but also the movable blade **60** itself can be easily formed.

[0041] The inclined surface **613** is inclined at an angle of 15° or more and 45° or less with respect to the second blade surface **612** when viewed from the radial direction Dr. Since the inclined surface **613** is inclined at an angle of 15° or more with respect to the second blade surface **612**, the generation of a Karman vortex can be suppressed with a high probability. Further, since the inclined surface **613** is inclined at an angle of 45° or less with respect to the second blade surface **612**, separation of the working fluid flowing along the inclined surface **613** can be suppressed.

[0042] The plurality of movable blades **60** increase the opening degree by rotating toward the reverse side to the rotational direction R of the rotor shaft **30**. The first blade surface **611** is located in front of the second blade surface **612** in the rotational direction R. Accordingly, in the movable blades **60**, when the opening degree is increased, the flow path between the movable blades **60** adjacent to each other at the trailing end is further widened by the inclined surfaces **613**. Thus, the head of the compressor can further be raised when the opening degree of the movable blades **60** is increased. Therefore, it is possible to further improve the performance of the compressor when the movable blades **60** are in the over-opened state.

Modifications

[0043] Next, an inlet guide vane **6A** of a modification example according to the disclosure will be described. Note that in the following description of the modification example, common components to those in the embodiment described above are denoted by the same reference signs in the drawings, and descriptions thereof are omitted. In the modification example, the rotational direction R of the rotor shaft **30** is different from that in the above-described embodiment.

[0044] As illustrated in FIG. 4, the rotational direction R of the rotor shaft **30** is reversed to that in the above-described embodiment. That is, a direction in which the rotor shaft **30** rotates in the circumferential direction Dc from the rear side to the front side in the circumferential direction Dc is reversed to that in the embodiment. In addition, a plurality of movable blades **60A** increase the opening degree by rotating toward the reverse side to the rotational direction R of the rotor shaft **30**. Thus, in the inlet guide vane **6A** of the modification example, the direction of rotation of the plurality of movable blades **60A** is different from that in the embodiment.

[0045] Even in the inlet guide vane **6A** of the modification example described above, similarly to the embodiment, the trailing end of the movable blade **60A** has an asymmetrical shape due to the inclined surface **613** when viewed from the radial direction Dr. As a result, it is possible to suppress the generation of a Karman vortex due to a fluid having passed through the inlet guide vane **6A**. That is, regardless of the rotational direction R of the rotor shaft **30**, the generation of a Karman vortex can be suppressed by the inclined surface **613**.

Other Embodiments

[0046] Although the embodiment of the disclosure has been described in detail with reference to the accompanying drawings, specific configurations are not limited to the embodiment, and include design modifications and the like without departing from the gist of the disclosure.

[0047] In the above-described embodiment, as an aspect of the geared compressor **1**, a so-called single-shaft two-stage configuration has been described as an example. However, the aspect of the geared compressor **1** is not limited thereto, and the geared compressor **1** may include a two-shaft four-stage configuration or a configuration of more shafts and more stages, in accordance with design, specifications, and the like.

[0048] The rotary machine of the disclosure is not limited to a geared compressor, and may be a single shaft multi-stage centrifugal compressor or the like in which the rotor shaft **30** is rotationally driven directly by an external drive source, or a gas turbine, a steam turbine, or the like.

[0049] In addition, the first blade surface **611** is not limited to being located in front of the second blade surface **612** in the rotational direction R. For example, in contrast to the embodiment and the modification example, the first blade surface **611** may be located behind the second blade surface **612** in the rotational direction R. In a case where the first blade surface **611** and the second blade surface **612** are arranged in this manner, the inclined surface **613** is inclined to face the second side Da2 in the axial direction Da so as to be opposed to the impeller **40** when the movable blade **60** is in the fully-closed state.

Supplementary Notes

[0050] The inlet guide vane **6**, **6A** and the rotary machine **1** described in the embodiment are understood as follows, for example. [0051] (1) An inlet guide vane **6**, **6A** according to a first aspect includes a plurality of movable blades **60**, **60A** disposed at intervals in a circumferential direction Dc of a rotor shaft **30** and rotatable about a center axis extending in a radial direction Dr of the rotor shaft **30**. Each of the movable blades **60**, **60A** includes a first blade surface **611** and a second blade surface **612** extending in a direction connecting a leading edge **61f** and a trailing edge **61r**, and an inclined surface **613** formed at a trailing end including the trailing edge **61r** and extending obliquely from the first blade surface **611** to the second blade surface **612** when viewed from the radial direction Dr. The inclined surface **613** is a flat surface whose cross-sectional shape viewed from the radial direction Dr is inclined so as to be connected, at an acute angle, to the second blade surface **612** at the trailing edge **61r**.

[0052] Accordingly, the trailing end of the movable blade **60**, **60A** has an asymmetrical shape due to the inclined surface **613** when viewed from the radial direction Dr. As a result, it is possible to suppress the generation of a Karman vortex due to a fluid having passed through the inlet guide vane **6**, **6A**. [0053] (2) An inlet guide vane **6**, **6A** according to a second aspect is the inlet guide vane **6**, **6A** according to (1), wherein each of the first blade surface **611** and the second blade surface **612** is formed of a flat surface whose cross-sectional shape viewed from the radial direction Dr forms a straight line in a direction connecting the leading edge **61f** and the trailing edge **61r**, and the first blade surface **611** and the second blade surface **612** are parallel to each other.

[0054] Accordingly, the movable blade has a linear shape. Thus, when the inlet guide vane **6**, **6A** is brought close to the fully-closed state, a gap between the movable blades **60**, **60A** adjacent to each other can be reduced. Accordingly, the leakage amount of the working fluid flowing through the gap can be reduced. As a result, even when the opening degree of the inlet guide vane **6**, **6A** is low, opening-degree characteristics can be improved. [0055] (3) An inlet guide vane **6**, **6A** according to a third aspect is the inlet guide vane **6**, **6A** according to (1) or (2), wherein the inclined surface **613** is inclined at an angle of 15° or more and 45° or less with respect to the second blade surface **612** when viewed from the radial direction Dr.

[0056] Since the inclined surface **613** is inclined at an angle of 15° or more with respect to the second blade surface **612**, the generation of a Karman vortex can be suppressed with a high probability. Further, since the inclined surface **613** is inclined at an angle of 45° or less with respect to the second blade surface **612**, the separation of the working fluid flowing along the inclined surface **613** can be suppressed. [0057] (4) An inlet guide vane **6**, **6A** according to a fourth aspect is the inlet guide vane **6**, **6A** according to any one of (1) to (3), wherein the plurality of movable blades **60**, **60A** increase an opening degree by rotating toward a reverse side to the rotational

direction R of the rotor shaft **30**, and the first blade surface **611** is located in front of the second blade surface **612** in the rotational direction R.

[0058] Accordingly, in the movable blades **60**, **60A**, when the opening degree is increased, the flow path between the movable blades **60**, **60A** adjacent to each other at the trailing end is further widened by the inclined surfaces **613**. Thus, the head of the compressor can further be raised when the opening degree of the movable blades **60**, **60A** is increased. Therefore, it is possible to further improve the performance of the compressor when the movable blades **60**, **60A** are in the over-opened state. [0059] (5) A rotary machine **1** according to a fifth aspect includes: a rotor **3** including a rotor shaft **30** extending in an axial direction Da in which an axis O extends with the axis O as a center of the rotor shaft **30**, and an impeller **40** fixed to the rotor shaft **30**; a casing **2** covering the rotor **3** and including a suction port **22a** through which a working fluid flows into the casing **2**; and the inlet guide vane **6**, **6A** according to any one of (1) to (4) disposed in the casing **2** on a first side Da1 in a direction of the axis O with respect to the impeller **40**.

[0060] While preferred embodiments of the disclosure have been described as above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the disclosure. The scope of the disclosure, therefore, is to be determined solely by the following claims.

Claims

1. An inlet guide vane comprising: a plurality of movable blades disposed at intervals in a circumferential direction of a rotor shaft, the movable blades being rotatable about a center axis extending in a radial direction of the rotor shaft, wherein each of the movable blades includes a first blade surface and a second blade surface extending in a direction connecting a leading edge and a trailing edge, and an inclined surface formed at a trailing end including the trailing edge, the inclined surface extending obliquely from the first blade surface to the second blade surface when viewed from the radial direction, and the inclined surface is a flat surface whose cross-sectional shape viewed from the radial direction is inclined so as to be connected, at an acute angle, to the second blade surface at the trailing edge.
 2. The inlet guide vane according to claim 1, wherein each of the first blade surface and the second blade surface is formed of a flat surface whose cross-sectional shape viewed from the radial direction forms a straight line in a direction connecting the leading edge and the trailing edge, and the first blade surface and the second blade surface are parallel to each other.
 3. The inlet guide vane according to claim 1, wherein the inclined surface is inclined at an angle of 15° or more and 45° or less with respect to the second blade surface when viewed from the radial direction.
 4. The inlet guide vane according to claim 1, wherein the plurality of movable blades increase an opening degree by rotating toward a reverse side to a rotational direction of the rotor shaft, and the first blade surface is located in front of the second blade surface in the rotational direction.
 5. A rotary machine comprising: a rotor including a rotor shaft extending in an axial direction in which an axis extends with the axis as a center, and an impeller fixed to the rotor shaft; a casing covering the rotor and including a suction port through which a working fluid flows into the casing; and the inlet guide vane according to claim 1 disposed in the casing on a first side in the axial direction with respect to the impeller.
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